OSPF

Course Code: CSC 3116 Course Title: Computer Networks

Dept. of Computer Science
Faculty of Science and Technology

Lecturer No:	Lab 8	**Week No:**		**Semester:**	Summer 22-23
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Lecture Outline

1. OSPF Theory
2. OSPF Configuration

Topic Heading..

Topic sub heading..

	EIGRP	OSPF
1	It supports maximum 255 routers in the network. However, the default is 100 routers. (highly scalable)	Supports unlimited number of routers
2	Fast convergence due to feasible successor	Fastest convergence speed due to the area concept
3	Cisco proprietary protocol and can be implemented only in Cisco routers.	Open standard protocol and can be implement in any router.
4	It calculates the metric In terms of bandwidth and delay (default).	It calculates the metric In terms of bandwidth only.
5	EIGRP works on DUAL(Diffusing Update Algorithm) Algorithm.	OSPF works on Dijkstra Algorithm.
6	It maintains the best route and some other alternative routes for each destination.	It maintains the best route in routing table and all routes in database table.
7	It is basically use for medium to lager size organization in the network [1].	It is basically use for lager size organization in the network [1].
8	Administrative distance 90	Administrative distance 110
9	Easy to implement	The implementation is complicated

OSPF Theory

OSPF Area

- An autonomous system (AS) is divided into one or more area.
- Each area is given an area ID
- An AS must have an area having ID 0 (zero) for multi-area OSPF. Such area is called backbone area.
- All areas of an AS must be connected to the backbone area.
- A router in an area exchanges routing information with the routers of its area only (by default)

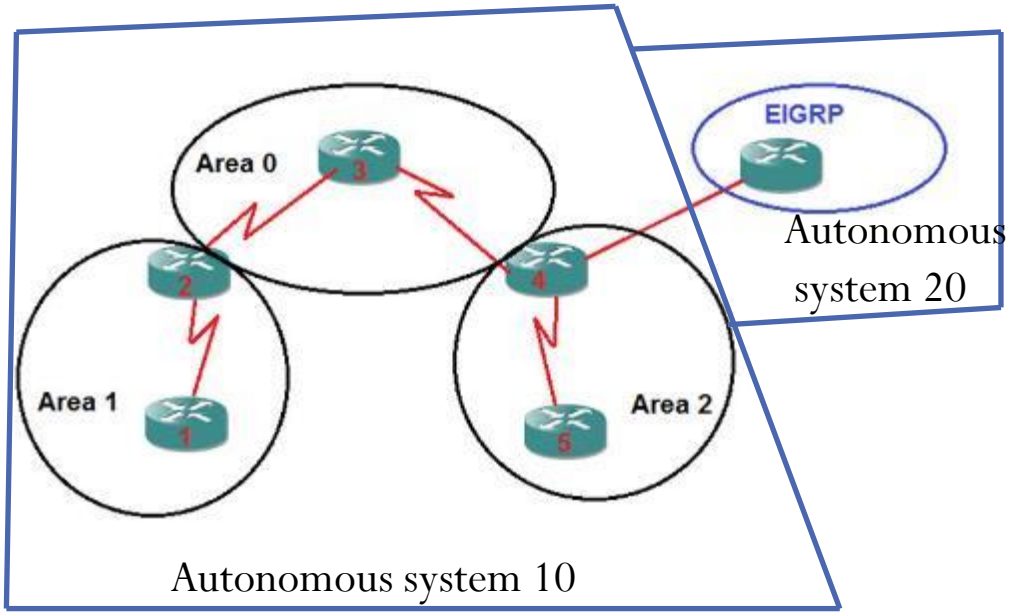


Fig. 1 Autonomous systems and area

OSPF Theory....

OSPF Routers

- Internal Router (IR): The router for which all its interface belong to one area. Router 1 and Router 5.
- Area Border Router (ABRs):The router that contains interfaces in more than one area. Router 2 and Router 4
- Backbone Router: The router that has all or at least one interface in Area 0. Router 3, Router 2 and Router 4.

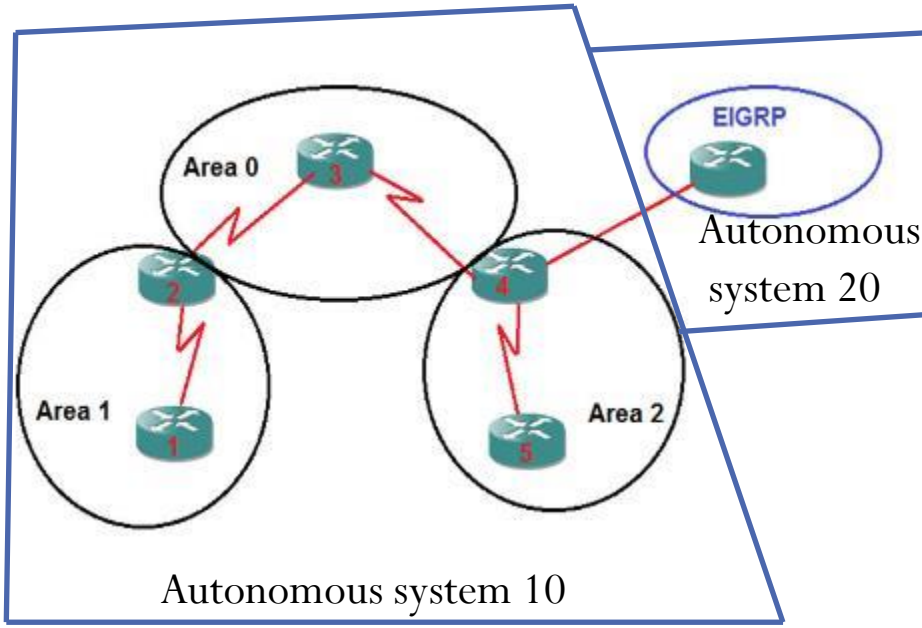


Fig. 1 Autonomous systems and area

- Autonomous System Boundary Router (ASBR): The routers with connection to a separate autonomous system. R4 in the example is connected to EIGRP [4].

OSPF Theory....

OSPF data structure and packets

- Link state advertisement (LSA)
  - A data structure with some specific information about the networks [2].
  - Depending on its type, it holds information about
    - a router's interfaces,
    - all routers attached to network,
    - summary routing information of an area,
    - all routers of an AS.
- Link state database (LSDB)
  - A collection of all LSAs known to a router
  - In a convergent network, all routers of a network have the same LSDB.

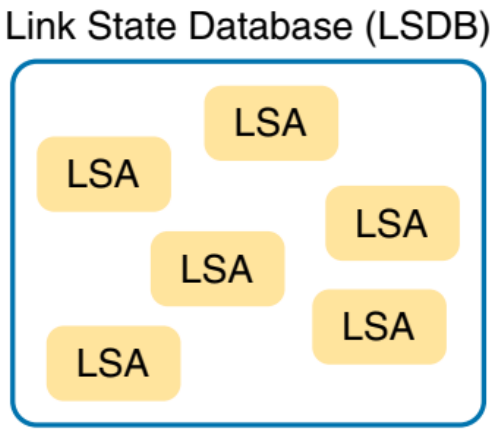


Fig. 2 LSA & LSDB relationship

OSPF Theory….

OSPF data structure and packets

- Hello

  Used to build and maintain neighbor relationships.
- DBD – Database Description

  List of LSAs contained in a LSDB. This packet type is circulated when two routers are initially exchanging their link-state databases.
- Link State Request (LSR)

  Used to request complete information about a link learned from another router.
- Link State Update (LSU)

  Used to send one or LSA(s)
- Links State Acknowledgement (LSAck)

  Used to acknowledge the reception of an LSA

OSPF Theory….

Neighbor discovery

Parameters need to be identical for two routers to become neighbors

- Network mask—*net mask of the sending router*
- Subnet number —derived using the subnet mask and each router's interface Internet Protocol (IP) address
- Area ID—*area ID of the sending interface*
- Hello interval—*how often Hello packets are transmitted*
- Dead interval—*how long to wait for Hello packets before terminating neighbor*
- Authentication type and password—*optional*
- Stub area flag—*specifies the type of stub area, if applicable* [3]

Hello packet contains all these information

OSPF Theory….

Neighbor discovery….

The scenario begins with the link down, so the routers have no knowledge of each other as OSPF neighbors

1. Link between R1 and R2 comes up
2. R1 sends the first Hello to multicast IP address 224. 0.0.5, so R2 learns of the existence of R1 as an OSPF router. At that point, R2 lists R1 as a neighbor, with an interim beginning state of init.

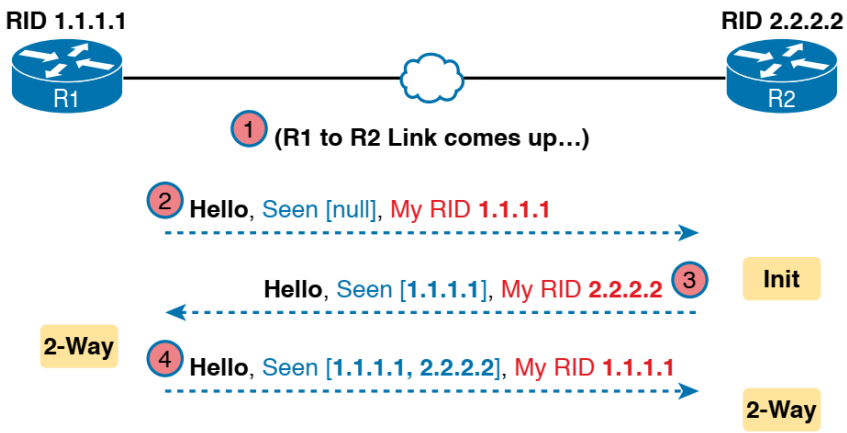


Fig. 3 Neighbor discovery

3. R2 sends back a Hello which tells R1 that R2 exists, and it allows R1 to move through the init state and quickly to a 2-way state.
4. R2 receives the next Hello from R1, and R2 can also move to a 2-way state [2]

OSPF Theory....

Router ID

- 32-bit unique dotted decimal number

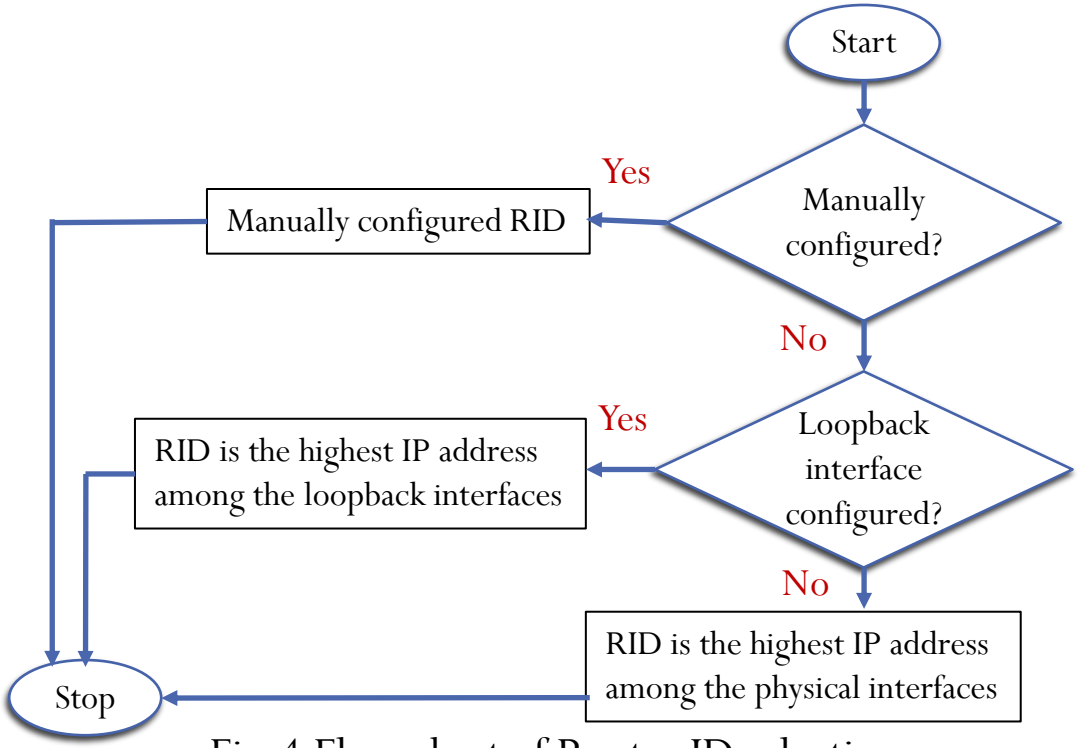


Fig. 4 Flow chart of Router ID selection

OSPF Theory....

DR and BDR

- Point-to-point network

  A network in an area connecting only two routers directly.
- Broadcast network

  A network in an area connecting more than two routers
- Designated router (DR)

  In a broad cast network, a router with the highest priority .

  If the priorities tie, the router having the highest RID

  All database exchange is done via DR
- Backup Designated router (BDR)

  In a broad cast network, a router with the second highest priority .

  If the priorities tie, the router having the second highest RID

  If the DR fails, the BDR takes over.

- DROTHER: The router which is neither DR nor BDR [2]

OSPF Theory....

DR and BDR....

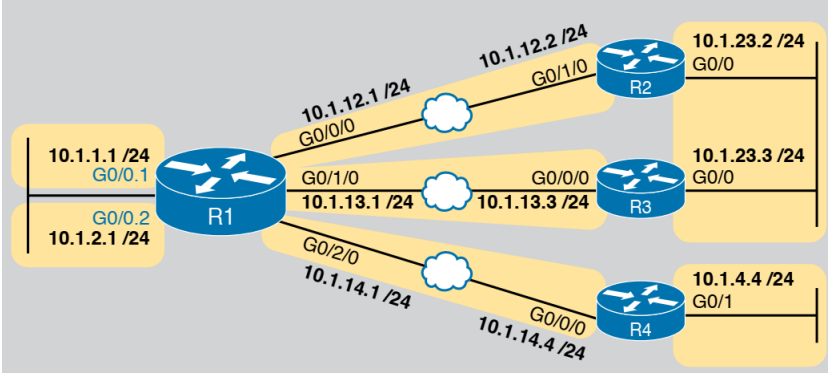


Fig. 5 point-to-point network

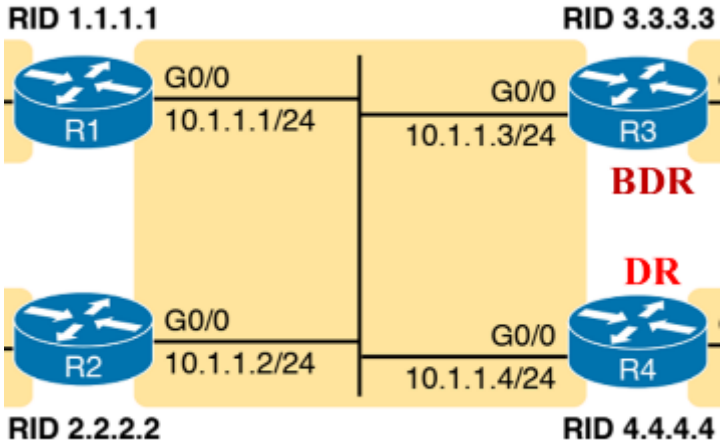


Fig. 6 Broadcast network , DR and BDR election

OSPF Theory....

DR and BDR....

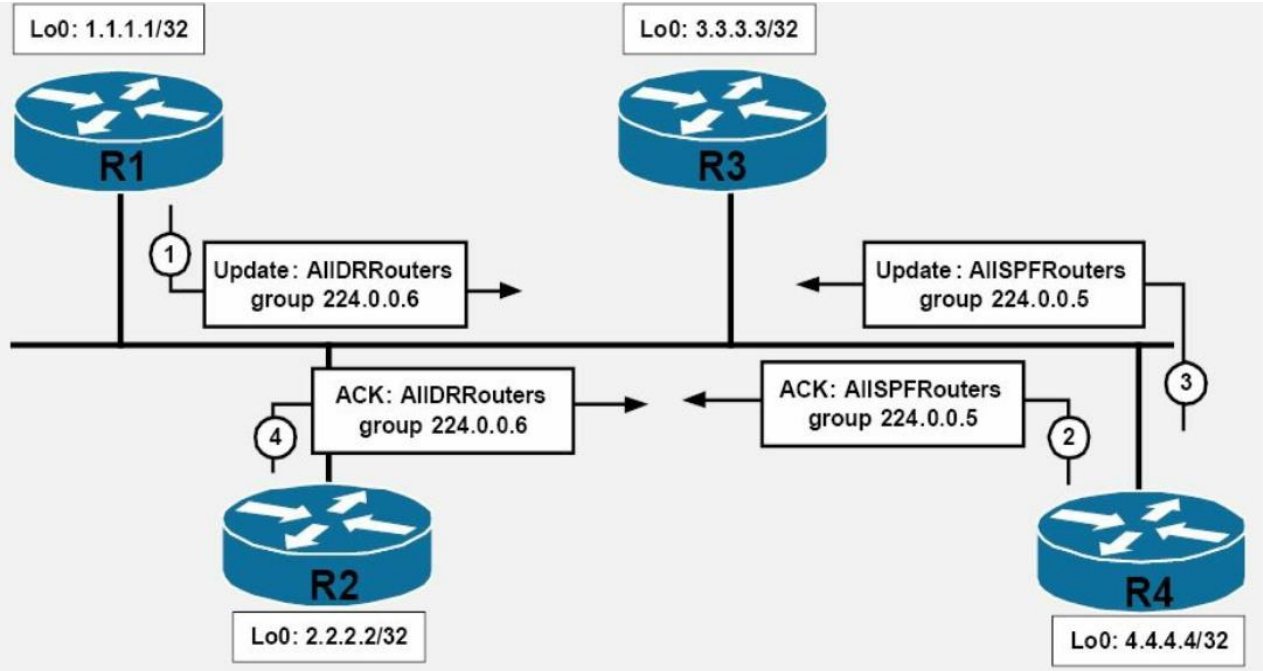


Fig. 7 Illustration of update exchange through DR and BDR

Configuration

Wildcard mask

- Used to specify a range of network addresses.
- Inverted subnet mask
- Used in EIGRP, OSPF and Access-List.

- How to get wildcard mask of an IP address?

  Subtract the subnet mask from 255.255.255.255

- What does each bit of a wildcard mask mean?

  0 : All IP address in the range must match the bit

  1 : Different IP address in the range can have different value in the bit position

Configuration....

Wildcard mask....

- Only 192.168.3.0

  All bits must match.

  WCM: All bits 0. (00000000.00000000.00000000.00000000)

  WCM: 0.0.0.0

- IP address range: 192.168.3.0 to 192.168.3.255

  Match first three block (24 bits) and fourth block can take any value

  WCM: 0.0.0.255

- IP address range: 192.168.3.4 to 192.168.3.13

  11000000.10101000.00000011.00000100

  11000000.10101000.00000011.00001101

First 28 bits same.

Match first 28 bits; make them all zero

Make rest of the bits 1

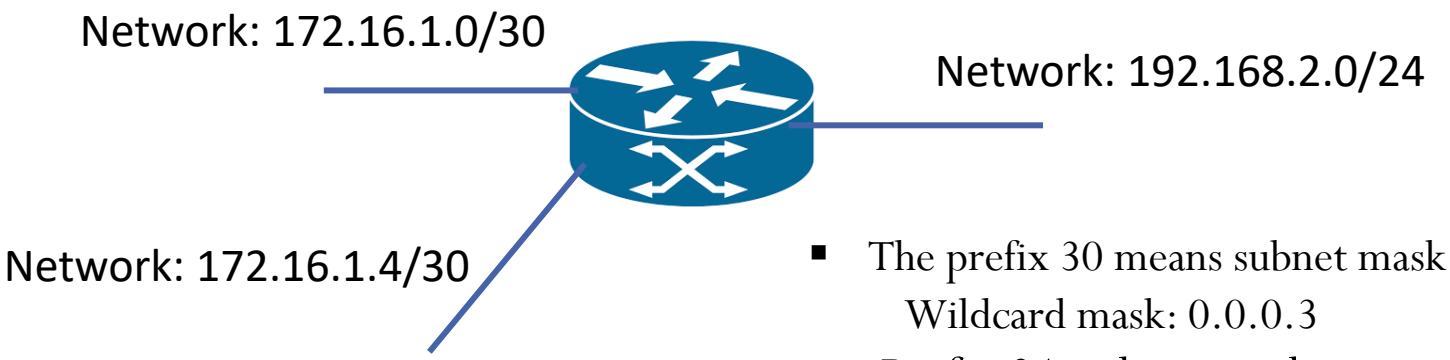
00000000.0000000.00000000.00001111

WCM: 0.0.0.15

Actual IP range: 192.168.3.0 to 192.168.3.15 under WCM: 0.0.0.15

Configuration....

Commands



- The prefix 30 means subnet mask: 255.255.255.252
  Wildcard mask: 0.0.0.3
- Prefix 24 indicates subnet mask: 255.255.255.0
  Wildcard mask: 0.0.0.255

OSPF Configuration

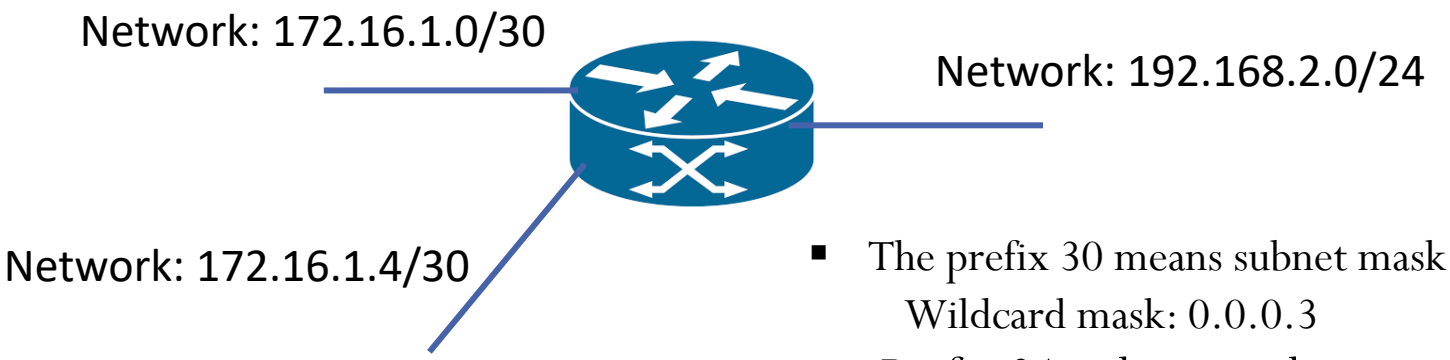
R1(config)# router ospf *process_id*

R1(config-router)#network *network_IP WCM* area *area_id*

Process ID is an integer. Not all routers need to have the same process ID.

Configuration….

Commands….

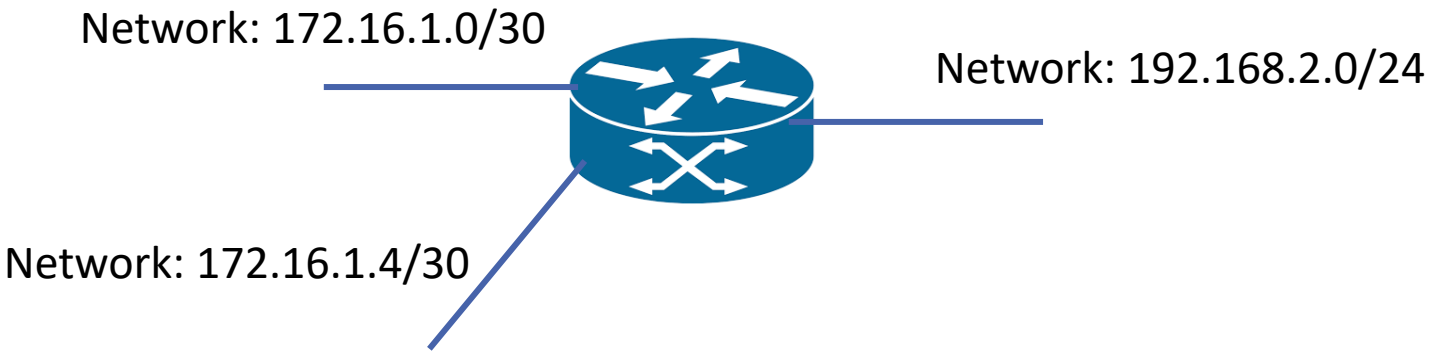


- The prefix 30 means subnet mask: 255.255.255.252
  Wildcard mask: 0.0.0.3
- Prefix 24 indicates subnet mask: 255.255.255.0
  Wildcard mask: 0.0.0.255

```
R1(config)# router ospf 10
R1(config)# router –id  1.1.1.1
R1(config-router)#network 192.168.2.0    0.0.0.255  area 0
R1(config-router)#network 172.16.1.0   0.0.0.3  area 0
R1(config-router)#network 172.16.1.4    0.0.0.3  area 0
```

Configuration….

Commands….




```
R1(config)# router ospf 10
R1(config-router)#network 192.168.2.0   0.0.0.255  area 0
R1(config-router)#network 172.16.1.0   0.0.0.255  area 0
```

ospf protocol will be used in any network of area 0 which is connected to the router, having network IP address with first three octets 172.16.1 [2]

Topic Heading..

Commands….

TABLE II Command for verifying configuration [2]

Command	Description
show ip ospf neighbor [*type number*]	Lists brief output about neighbors, identified by neighbor router ID, including current state, with one line per neighbor; optionally, limits the output to neighbors on the listed interface.
show ip ospf neighbor *neighbor-ID*	Lists the same output as the **show ip ospf neighbor** detail command, but only for the listed neighbor (by neighbor RID).
show ip ospf database	Lists a summary of the LSAs in the database, with one line of output per LSA. It is organized by LSA type (first type 1, then type 2, and so on).
show ip route	Lists all IPv4 routes.
show ip route ospf	Lists routes in the routing table learned by OSPF.
show ip route *ip-address mask*	Shows a detailed description of the route for the listed subnet/mask.
clear ip ospf process	Resets the OSPF process, resetting all neighbor relationships and also causing the process to make a choice of OSPF RID.

References

[1] Computer Networking Class, https://computernetworkingclass.blogspot.com… /2016/08/comparison-between-rip-eigrp-igrp-and.html, [Accessed: May 2, 2020].

[2] W. Odom, Official Cert Gudie CCNA 200-301 Volume 1, Pearson Education, Inc., USA, 2020, pp. 449-497.

[3] J. Macfarlane, *Network Routing Basics*, Wiley Publications. Inc., USA, 2006, pp. 254.

[4] OSPF, "http://www.certiology.com/cisco-certifications/ccna/ccna-routing-and-switching/free-cisco-ccna-study-guide/ospf.html, [Accessed: May 2, 2020]..

Books

1. Official Cert Guide CCNA 200-301 , vol. 1, *W. Odom*, Cisco Press, First Edition, 2019, USA.

2. CCNA Routing and Switching, *T. Lammle*, John Wily & Sons, Second Edition, 2016, USA.